

CIS 371 Web Application Programming

JS|TS Promise

Handling Asynchronous Results



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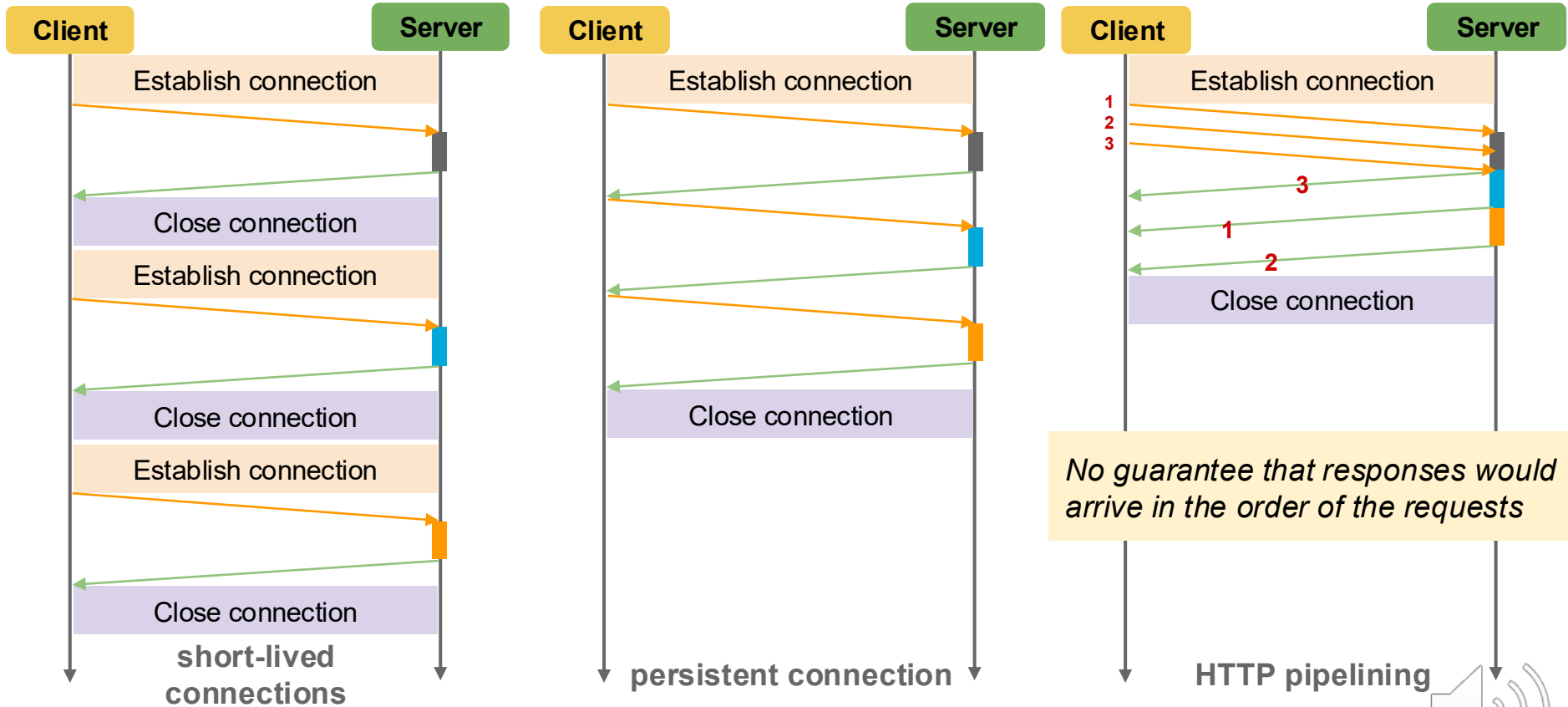


Topics

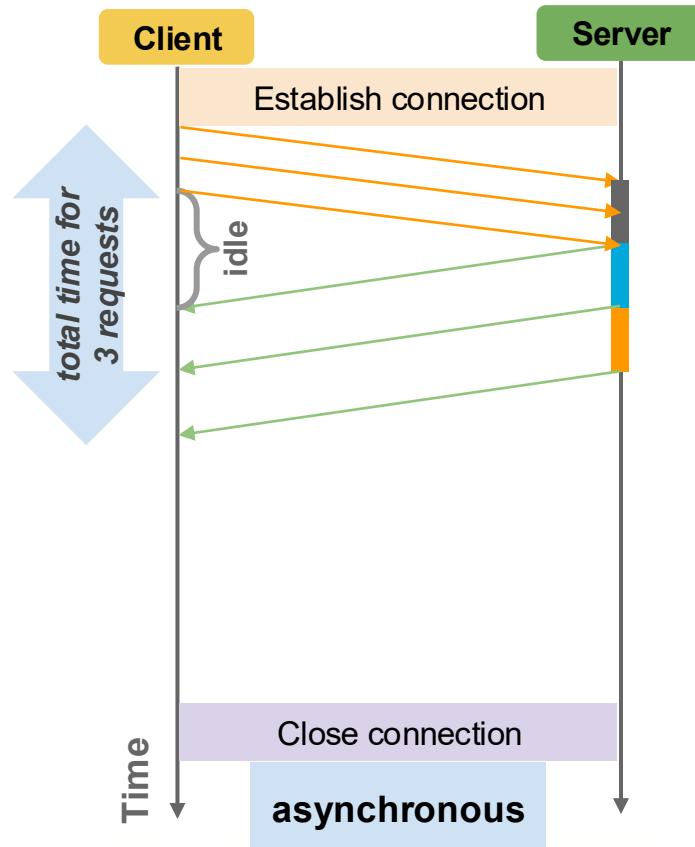
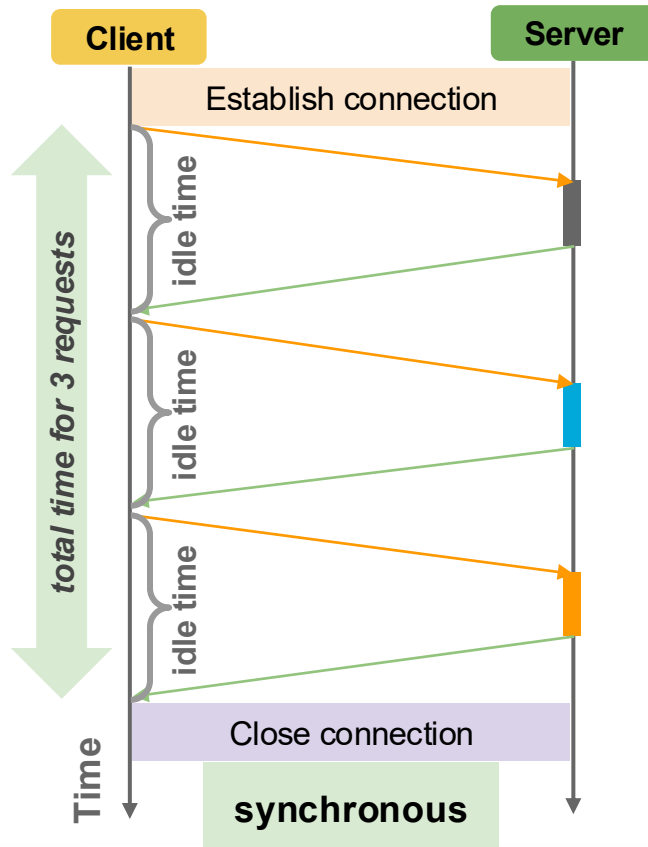
- Client/Server Communication
 - Synchronous
 - Asynchronous
- Callback functions (for handling asynchronous events)
- Promise

Reference: Promise Documentation (@ MDN)

Client/Server: HTTP Requests & Responses

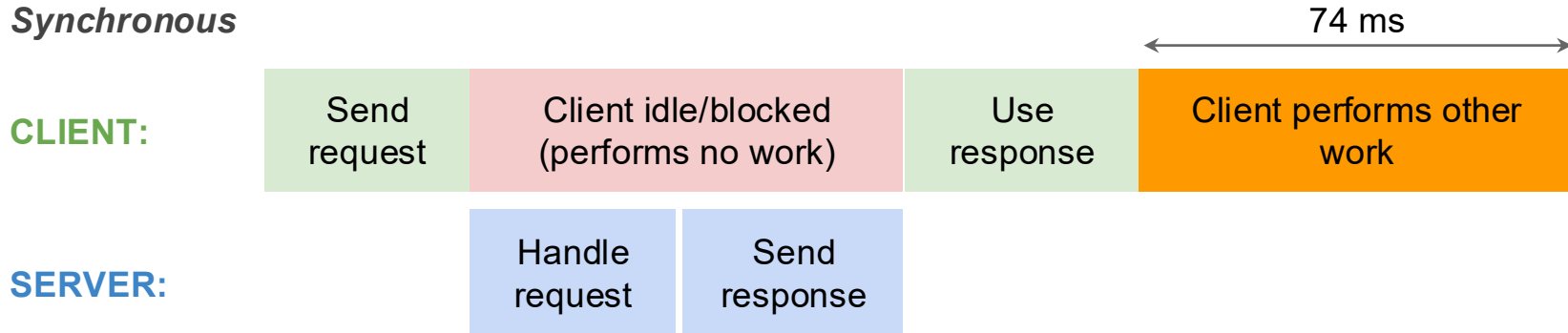


Client/Server: HTTP Requests & Responses

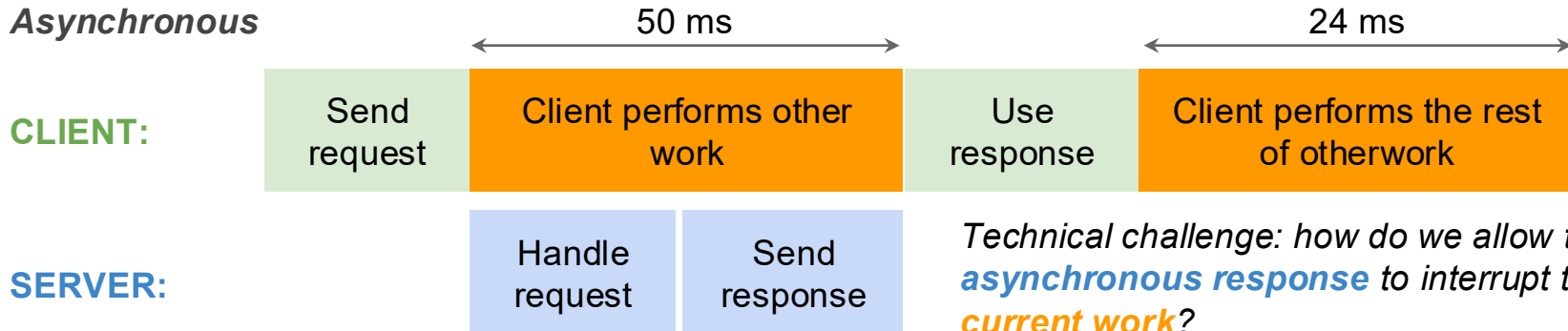


Synchronous vs. Asynchronous Requests

Synchronous



Asynchronous



Technical challenge: how do we allow the **server asynchronous response** to interrupt the **client current work**?



Sending Requests: easy
Receiving Async Responses: requires extra setup

Callback Actions (JS Callback Functions)

You are number 17 in line.....



Would you like us to call you back?

1-888-I-CAN-HELP

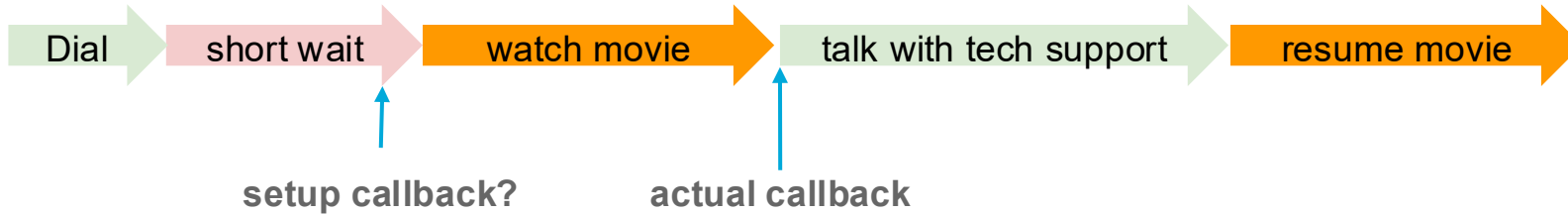




Option #1: without callback



Option #2: with callback



Synchronous Call (in code)

555-4321



1-888-I-CAN-HELP



Dial

connect & extremely long wait

talk with tech support

watch movie

```
dial("888-I-CAN-HELP");  
connect_and_long_wait();  
talk_with_tech();  
watch_movie();
```

*Order of execution = order of
line of code*

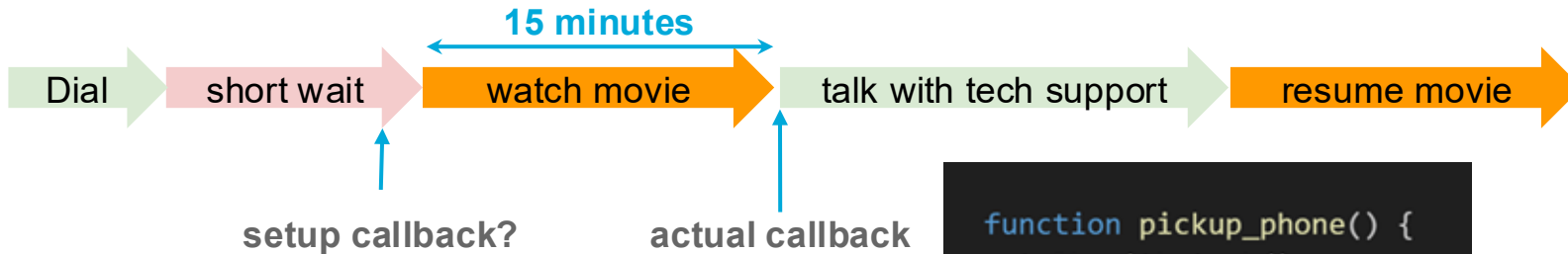
**Async: “out-of-order” execution
(Order of execution $\neq$ order of line of code)**

Async Phone Calls with Callback (in code)

555-4321



1-888-I-CAN-HELP



```
dial("888-I-CAN-HELP");  
setup_cb("555-4321", pickup_phone);  
watch_movie();
```

```
function pickup_phone() {  
  talk_with_tech();  
}
```

15 mins later

Asynchronous (incoming call) while you're watching movie

Callback fns (Fat Arrow)

```
function pickup_phone() {  
  talk_with_tech();  
}  
dial("888-I-CAN-HELP");  
setup_cb("555-4321", pickup_phone);  
watch_movie();
```

named function

```
dial("888-I-CAN-HELP"); 1  
setup_cb("555-4321", () => { 2  
  // 15 min later  
  talk_with_tech(); 4  
});  
watch_movie(); 3 5
```

fat arrow



What is the execution order?

Async: order of execution $\neq$ order of line of code

555-4321

1-888-I-CAN-HELP



15 minutes

Dial

short wait

watch movie

talk with tech support

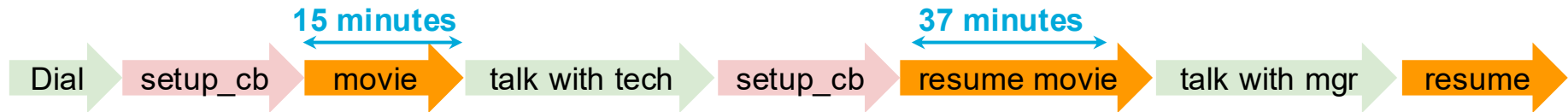
resume movie

setup callback?

actual callback

```
// start dialing ...  
dial("888-I-CAN-HELP"); 1  
// call me back @ 555-4321  
// then hangup to watch movie  
setup_cb("555-4321", () => { 2  
    // 15-min later  
    talk_with_tech(); 4  
});  
// watch it NOW!!!  
watch_movie(); 3 5
```

**Tech: “But, you have to talk with my manager”
(Nested Callback)**



```
dial("888-I-CAN-HELP"); 1
setup_cb("555-4321", () => { 2
  // 15-min later
  // Talk with tech 4
  setup_cb("555-4321", () => { 5
    // 37-min later
    // Talk with manager 7
  });
});
watch_movie(); 3 6 8
```

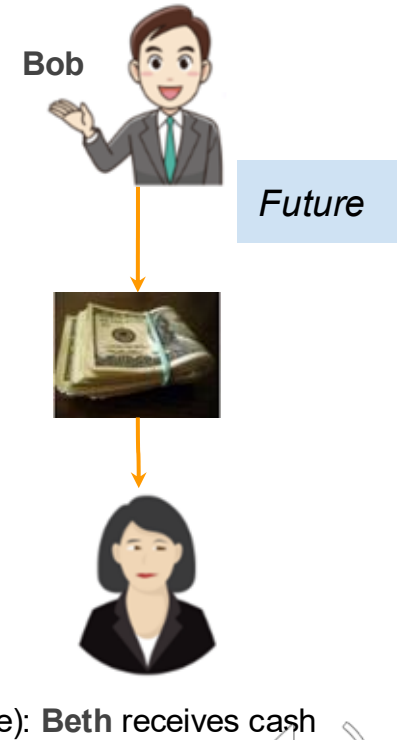
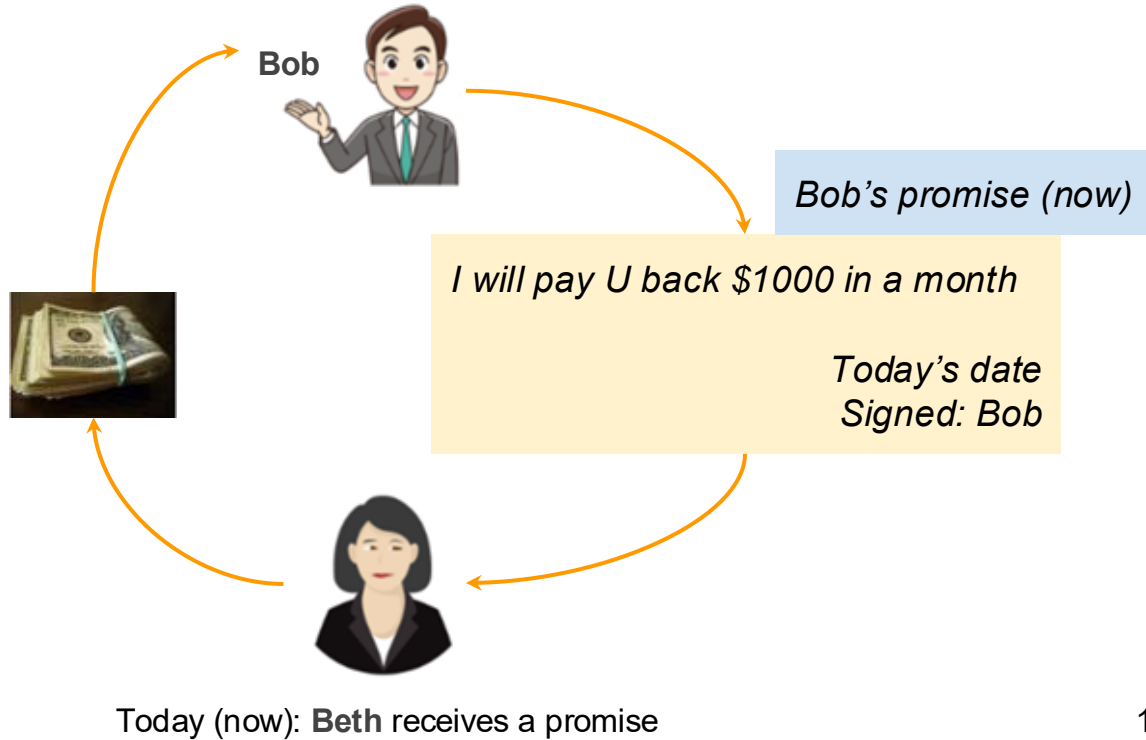
Nested
callbacks

How to Initiate Async HTTP Requests?

- `fetch()` function
 - Native in browser
 - NPM `node-fetch`
- Axios library
- Both `fetch()` and `axios()` use JS Promise

IOU = I owe you note
Promise to pay debt/loan

Borrowing Money: Promise Now, Pay Later



A promise = now confirmation of future action(s)
**A JS promise = a “now” object representing data
which will become available in the future**

Promise Example

```
function nthPrime(nth: number): Promise<number> {  
  // work takes 10 seconds  
  return Promise.resolve(____);  
}
```

```
console.log("Start");  
const prom = nthPrime(500);  
prom.then((pr: number) => {  
  console.log("The 500th prime is", pr);  
});  
doMoreWork();
```

Start
Partial output of doMoreWork()
After 10 seconds
The 500th prime is 3571
More output from doMoreWork()

Compare the order of execution

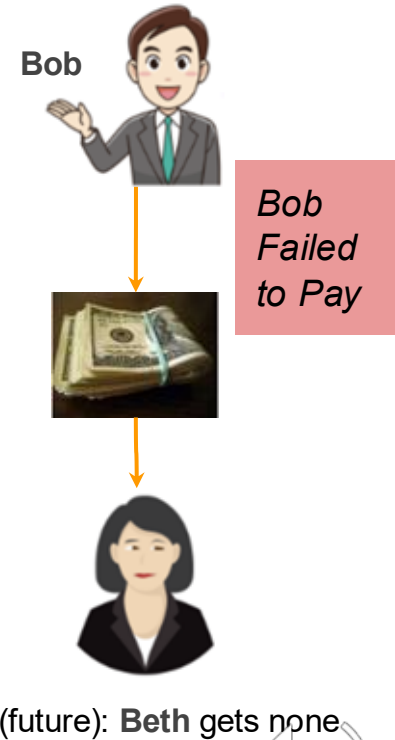
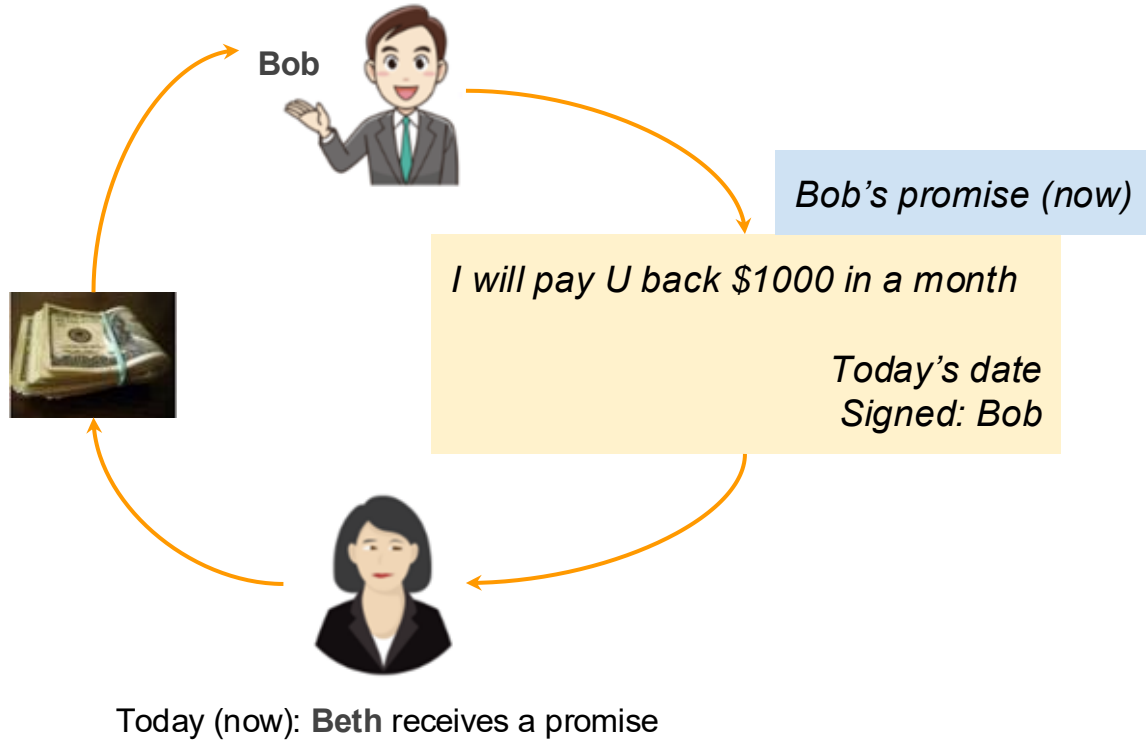
```
function nthPrimeNow(nth: number): number {  
  // work takes 10 seconds  
  return ____;  
}
```

```
console.log("Start");  
const pr = nthPrimeNow(500);  
console.log("The 500th prime is", pr);  
doMoreWork();
```

Start
After 10 seconds
The 500th prime is 3571
Output of doMoreWork()

Loan is either paid-off or defaulted
Promise is either resolved or rejected

Borrowing Money: Promise Now, **Never** Pay



Promise settlement: resolve() or reject()

```
function nthPrime(nth: number): Promise<number> {  
  if (nth < 100_000) {  
    // assume prime calculation takes 10 seconds  
    return Promise.resolve(a_prime_number_here);  
  } else return Promise.reject("Can't compute prime");  
}
```

```
console.log("Start");  
nthPrime(500).then((pr: number) => {  
  console.log("Prime is", pr);  
});  
.catch((err:any) => {  
  console.log("Rejected", err);  
});  
console.log("Here");
```

```
# Watch for order of execution  
Start  
Here  
# if the promise is resolved  
# After 10 seconds ...  
Prime is 3571  
# if the promise is rejected  
Rejected Can't compute prime
```

JS Promise

- Basic methods: `then()`, `catch()`, `finally()`
- Basic static functions
 - `Promise.resolve()`
 - `Promise.reject()`
- Advanced utilities for handling multiple promises
 - `Promise.all(array)`: wait until all the promises in the array are resolved
 - `Promise.allSettled(array)`: wait until all the promises in the array are either resolved or rejected
 - `Promise.any(array)`: wait until ONE of the promises in the array is resolved
 - `Promise.race(array)`: wait until ONE of the promises in the array is either resolved or rejected

then-able chains

Then and then and then and ...

```
function nthPrime(nth: number): Promise<number> {  
    // more code here  
    return Promise.____;  
}
```

```
function toRomanNumeral(inputNum: number): string {  
    // conversion to Roman numeral  
    return ____;  
}
```

Return from a then() becomes a Promise to the next then() inline

```
nthPrime(500)  
    .then((p: number): string => {  
        return toRomanNumeral(p);  
    })  
    .then((rome: string) => {  
        console.log(`Prime in roman numeral ${rome}`);  
    });
```

```
// After 1-line return elimination  
nthPrime(500)  
    .then((p: number): string => toRomanNumeral(p))  
    .then((rome: string) => {  
        console.log(`Prime in roman numeral ${rome}`);  
    });
```

Demo

Then and then and ... (promise “unpacked”)

```
function nthPrime(nth: number): Promise<number> {  
  // more code here  
  return Promise.____;  
}
```

```
nthPrime(500)  
  .then((p: number): Promise<string> => promNum(p))  
  .then((rome: string) => {  
    // "unpacked"!!!  
    console.log(`Prime in roman numeral ${rome}`);  
  });
```

```
function promNum(inputNum: number): Promise<string> {  
  // conversion to Roman numeral  
  return Promise.____;  
}
```

Demo



Online Pizza Order & 3rd party payment



Online Pizza Order (code setup)

```
function orderPizza(__): Promise<PizzaOrder> {  
  return Promise.resolve(__);  
}
```

PizzaHat

```
function makePizza(__): Promise<PizzaBox> {  
  return Promise.resolve(__);  
}
```

PizzaHat

```
type PizzaOrder = {  
  crustStyle: "Classic" | "ThinCrust" | "HandTossed";  
  size: number;  
  toppings: Array<Topping>;  
  customerName: string;  
  price: number;  
};
```

```
type PizzaBox = {  
  customerName: string;  
  inStorePickup: boolean;  
};
```

```
type ProofOfPlay = {  
  payer: string;  
  payee: string;  
  amount: number;  
  transactionDate: string;  
};
```

```
function playWithPal(name: string, payAmt: number): Promise<ProofOfPlay> {  
  return Promise.resolve(__);  
}
```

PlayPal

Online Pizza Order (chaining)

```
function orderPizza(__): Promise<PizzaOrder> {  
  return Promise.resolve(__);  
}
```

PizzaHat

```
function makePizza(__): Promise<PizzaBox> {  
  return Promise.resolve(__);  
}
```

PizzaHat

```
function playWithPal(name: string, payAmt: number): Promise<ProofOfPlay> {  
  return Promise.resolve(__);  
}
```

PlayPal

```
orderPizza(__)  
  .then((ord: PizzaOrder) => playWithPal(__, __))  
  .then((proof: ProofOfPlay) => makePizza(__))  
  .then((box: PizzaBox) => {  
    console.log("Open the box and enjoy!");  
  })  
  .catch((err: any) => {  
    console.error("Can't complete order");  
  });
```


Promise: with finally

```
function nthPrime(nth: number): Promise<number> {  
  // more code here  
  return Promise.____;  
}
```

```
console.log("Start");  
nthPrime(500).then((pr: number) => {  
  console.log("Prime is", pr);  
});  
doMoreWork();
```

Start
Partial output of doMoreWork()
After 10 seconds
Prime is 3571
More output from doMoreWork()

```
console.log("Start");  
nthPrime(500)  
  .then((pr: number) => {  
    console.log("Prime is", pr);  
  })  
  .finally(() => {  
    doMoreWork();  
  });
```

Start
After 10 seconds
Prime is 3571
Output of doMoreWork()

The finally method is used to specify a block of code that will run after the promise is settled, regardless of whether it was resolved or rejected.

Promise: put them all together

```
work_with_promise(____, ____, ____)  
  .then((arg: type1): type2 => {  
    // more code here  
    return ____;  
  })  
  .then((arg: type2): type3 => {  
    // more code here  
    return ____;  
  })  
  .then((arg: type2): type3 => {  
    // more code here  
    return ____;  
  })  
  /* more chain of .then here */  
  .catch((err: any) => {  
    // Error handling code here  
  })  
  .finally(() => {  
    // Overall "cleanup" code here  
  });
```

Any Promise.reject() here will be caught by

Promise.reject() skips then-chain until it finds a .catch

async & await

Async functions



What is the difference between promise functions with and without the `async` keyword?

```
function nthPrime(nth: number): Promise<number> {  
  let thePrime: number;  
  // more code here  
  return Promise.resolve(thePrime);  
}
```

The `async` keyword makes asynchronous functions look and behave more like synchronously way by

- *removing the need for explicit promise creation.*
- *using the `await` keyword to pause the `async` function execution*

```
async function nthPrime(nth: number): Promise<number> {  
  let thePrime: number;  
  // more code here  
  return thePrime; // Promise.resolve() is not required  
}
```

```
const nthPrime = async (nth: number): Promise<number> => {  
  let thePrime: number;  
  // more code here  
  return thePrime; // Promise.resolve() is not required  
};
```

await: rewrite in synchronous style

```
orderPizza(____)
  .then((ord: PizzaOrder) => playWithPal(__, __))
  .then((proof: ProofOfPlay) => makePizza(____))
  .then((box: PizzaBox) => {
    console.log("Open the box and enjoy!");
  })
  .catch((err: any) => {
    console.error("Can't complete order");
  });
```

Await can only be used inside async functions

```
const doPizza = async (): Promise<void> => {
  try {
    const ord: PizzaOrder = await orderPizza(____);
    const proof: ProofOfPlay = await playWithPal(__, __);
    const box: PizzaBox = await makePizza(____);
    console.log("Open the box and enjoy!");
  } catch (err) {
    console.error("Can't complete order");
  }
};
```